



Social Media Analysis for Sentiment Mining

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ASHLY PETER ARUJA (30116535)

Supervised By

Abigail Peters

Abstract

In the digital age, customer sentiment plays a critical role in shaping brand perception, especially for globally recognised companies like Nike. With the proliferation of review platforms and social media, understanding how consumers feel—and why—has become a strategic imperative. This study aimed to explore public sentiment toward Nike by analysing customer-generated reviews from Trustpilot using natural language processing (NLP) techniques. The research focused on uncovering the emotional tone of reviews, identifying key themes in customer discourse, and examining how sentiment varied across different topics.

A dataset of English-language Trustpilot reviews was collected and preprocessed for analysis. Sentiment classification was performed using the VADER tool, categorising reviews into positive, negative, or neutral sentiments. Latent Dirichlet Allocation (LDA) was used to extract underlying topics from the review corpus. The study then cross-referenced sentiment labels with dominant topics to reveal sentiment-topic relationships.

The results indicated that sentiment was nearly evenly divided between positive and negative, with strong emotional polarisation. Five key topics were identified: Product Quality & Purchase, Shipping & Delivery, Customer Support Issues, Returns & Refunds, and Website/App Experience. While product-related feedback was largely positive, service-related topics were more negatively perceived. This thematic-sentiment breakdown revealed specific operational areas for improvement.

The findings offer both academic and practical contributions by demonstrating how interpretable NLP models can generate actionable brand insights. The study concludes that while Nike maintains strong brand equity through its products, attention must be given to post-purchase service, refund processes, and digital platform usability. Recommendations for future work include the use of deep learning models and the inclusion of multi-platform, multilingual datasets.

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Chapter 1: Introduction

1.1 Background and Context

In today's digital landscape, where a brand's image is shaped as much by virtual interactions as by product performance, it is vital for businesses to stay responsive to public opinion. More and more, consumers participate in electronic word-of-mouth (eWOM), sharing their thoughts and personal encounters on platforms like social media, review websites, and discussion forums.

These digital spaces have emerged as key instruments for shaping brand identity and influencing how companies are perceived (Cheung and Thadani, 2012). Online reviews play a critical role by providing candid, first-hand insights into customer satisfaction, complaints, and expectations. Platforms such as Trustpilot have emerged as popular outlets for consumers to evaluate brands based on personal experiences. Unlike structured rating systems, review sites offer rich textual data that allows for deeper analysis of consumer sentiment. These reviews, when mined properly, reveal not just how customers rate a brand but why they feel a certain way. This distinction is vital in understanding the true drivers of customer satisfaction and dissatisfaction.

Nike, a leading global brand in athletic wear, maintains a substantial online presence and attracts significant public attention — both positive and negative. As a dominant force in the sportswear industry, Nike's reputation is subject to continuous scrutiny from a large, diverse customer base. Its products, services, delivery systems, and customer support are routinely discussed in public forums. This makes Nike an ideal subject for sentiment-based research aimed at uncovering the nuanced opinions of customers worldwide.

1.2 Rationale for Studying Nike

Nike, a globally renowned leader in athletic apparel, is known for the company's extensive global reach and strong marketing presence. Despite its widespread popularity, Nike frequently attracts both praise and criticism across various customer touchpoints. With a significant portion of its business now occurring online, Nike faces increased scrutiny regarding delivery times, customer service, product quality, and return policies.

This study was also personally motivated by the researcher's own brand affinity — Nike has long been my favourite sneaker brand due to its innovation, design, and presence in sports culture. As a loyal customer and admirer of the brand, I was particularly interested in exploring how other consumers experience Nike's products and services and whether the broader public perception aligns with my own.

Although review scores offer a broad overview of customer satisfaction, they are limited in capturing the deeper meaning found in written feedback. Therefore, it's important to examine unstructured customer review text through Natural Language Processing (NLP) methods that enable the extraction of meaningful emotional tones, the recognition of common concerns, and the identification of patterns in user responses (Cambria et al., 2013).

This research utilises NLP techniques to carry out an in-depth sentiment as well as topic evaluation of Nike user reviews collected from Trustpilot. The aim is not only to gauge the overall sentiment but also to identify the specific factors that drive satisfaction or dissatisfaction.

1.3 Aims and Objectives

The central focus of this dissertation is to explore public sentiment regarding Nike by analyzing customer-generated reviews on Trustpilot. Using Natural Language Processing (NLP) methods along with supervised and unsupervised ML techniques, the study aims to identify key patterns within customer opinions and assess how sentiment shifts across topics, time, and region.

The main objectives of the research are as follows:

1. To gather and prepare a dataset of Nike customer reviews obtained from Trustpilot.
2. To carry out sentiment evaluation and categorise each review into positive, negative, or neutral categories.
3. To implement topic modelling (via LDA) in order to find key recurring subjects in customer feedback.
4. To examine the relationship between sentiment and topic.
5. To analyse changes in sentiment and themes across different time periods and geographic areas.
6. To generate visual insights that can inform Nike's brand strategy and customer experience initiatives.

1.4 Research Questions

To support this inquiry, the study sets out to answer the following questions:

- Which types of sentiment are most frequently expressed by Nike customers on Trustpilot?
- Which topics or themes emerge most frequently in these reviews?
- How does sentiment vary across different topics and regions?
- How does customer sentiment shift or evolve over time?

1.5 Significance of the Study

This research makes contributions in both academic and practical domains of brand analytics. From an academic perspective, it illustrates the application of machine learning and NLP methods for analysing real-world customer reviews with respect to sentiment and themes. It underscores the value of review mining for gaining insights into customer viewpoints beyond standard numeric ratings (Pang and Lee, 2008).

From a practical angle, the findings provide actionable knowledge for brand managers, marketing professionals, and customer experience teams. Grasping the overall tone and thematic elements of customer responses can support Nike — as well as other companies — in resolving dissatisfaction more efficiently, enhancing brand image, and tailoring services to meet shifting customer expectations (Liu, 2012).

By analysing how sentiments vary by topic, region, and time, the study equips decision-makers with granular insights into customer experience trends. Furthermore, this research reflects the

growing importance of unstructured data analysis in strategic brand management, particularly in an era where digital-first customer interactions dominate the retail landscape.

1.6 Scope of the Study

This research centres on the analysis of unstructured textual reviews about Nike, collected from the Trustpilot platform. The dataset comprises English-language reviews submitted by customers across various geographic regions. The scope is limited to text-based content, excluding images, star ratings, or video material. Reviews were gathered over a defined time period to ensure a representative sample of customer sentiment.

The analysis applies two core natural language processing (NLP) techniques: sentiment evaluation along with thematic modelling. The sentiment analysis employed the VADER tool, a lexicon-based model designed for short, informal texts (Hutto and Gilbert, 2014), while topic modelling used the Latent Dirichlet Allocation (LDA) algorithm to identify recurring themes present in the review content. These methods were chosen for their compatibility with user-generated content and their ability to extract both emotional tone and underlying topics from free-text data.

1.7 Limitations of the Study

Although the research offers meaningful contributions to understanding consumer sentiment, it is important to recognise its limitations. The dataset is restricted to reviews collected from a single platform (Trustpilot), which may not fully reflect customer experiences shared on other platforms such as social media, brand websites, or global retail marketplaces. Moreover, the analysis was limited to English-language reviews, which may have left out perspectives from users who speak languages other than English, thereby reducing the range of cultural and linguistic viewpoints represented. The analysis was also confined to textual data, omitting structured metadata like ratings or reviewer demographics (e.g., age, gender, or geographic region).

Additionally, relying on automated NLP tools may result in missed nuances such as sarcasm, slang, ambiguous phrasing, and cultural nuance. These challenges are well documented in short-text sentiment analysis, where informal language, implicit meaning, and limited context reduce the effectiveness of traditional rule-based models (Maynard and Funk, 2011).

Chapter 2: Literature Review

2.1 Introduction

Sentiment analysis, sometimes called opinion mining, is a Natural Language Processing (NLP) method that is used to detect, extract, and measure emotional meaning from written data. As digital communication continues to grow—particularly through customer reviews, social platforms, and online discussions—sentiment analysis has become increasingly vital in understanding public perception at scale. It provides organisations with a way to systematically monitor consumer attitudes, improve service offerings, and track changes in brand perception over time.

At its foundation, sentiment analysis includes categorising text based on emotional tone into labels like positive, negative, or neutral. More advanced approaches go further by assessing the intensity of sentiment or identifying sentiment directed at specific aspects of a product or service (e.g., pricing, delivery, or customer support). These insights allow companies to pinpoint not just how consumers feel, but also what exactly they feel strongly about.

The process operates at multiple levels: review-level sentiment analysis examines the overall emotional tone of the entire text, sentence-level techniques segment sentiment within specific sentences, and aspect-level analysis links opinions to particular features. This layered approach is particularly useful in customer review analysis, where a single comment may praise one aspect of a brand while criticising another.

However, sentiment analysis also faces several challenges. These include handling sarcasm, interpreting slang, resolving ambiguity in language, and detecting mixed sentiments in a single review. One of the field's core difficulties is capturing the nuance of human emotion and intention from unstructured, often informal text (Mejova, 2009).

Despite these limitations, sentiment analysis is continually advancing through the use of machine learning, deep learning, and customised lexicons. Its importance in brand monitoring, user interaction, and market analysis makes it an essential resource for companies aiming to stay attuned to consumer feedback in an increasingly digital environment.

2.2 Techniques and Tools

Sentiment analysis relies on a diverse range of methods and tools to extract, classify, and evaluate emotional tone from text. These techniques are generally divided into two main types: rule-based systems and models driven by machine learning. The first type makes use of fixed linguistic rules and sentiment dictionaries, while the second relies on statistical algorithms trained on large text datasets to find patterns in language.

Rule-Based Approaches

Rule-based methods depend on predefined sentiment dictionaries that associate emotional values with specific terms. For instance, sentiment tools like VADER (Valence Aware Dictionary and sEntiment Reasoner) and TextBlob use tagged word banks that link terms with polarity indicators to gauge overall emotional tone. These tools work particularly well for analysing social media posts and product feedback, where informal phrasing is typical. VADER is favoured because of how effectively it manages punctuation, capitalisation, negation, and symbols such as emojis—making it well-suited for short-form opinion-based content. Research indicates that VADER delivers better results than many competing tools when analysing brief and casual writing formats, including those posted on platforms like social media or found in consumer feedback (Das et al., 2021).

However, rule-based techniques struggle with more nuanced interpretation, such as sarcasm, idioms, or complex sentence structures. This challenge often leads to hybrid systems that blend statistical learning with dictionary-driven approaches.

Machine Learning-Based Approaches

ML-based methods improve sentiment classification by analysing labelled datasets. Common supervised models, including algorithms like Naive Bayes, Support Vector Machines (SVM), and Logistic Regression, depend on training sets in which individual reviews are annotated with corresponding sentiment labels. These systems are trained to link distinctive linguistic patterns—such as word frequency, part-of-speech tags, or n-grams—to specific sentiment classifications.

Recent developments have introduced advanced neural models such as LSTM (Long Short-Term Memory) networks and transformer architectures like BERT (Bidirectional Encoder Representations from Transformers). These models offer superior contextual understanding and notably enhance sentiment classification performance, especially in complex or lengthy texts. BERT is initially trained on extensive datasets using masked word prediction and next-sentence training objectives, which enables it to grasp bidirectional context and deliver strong results in various natural language processing tasks (Devlin et al., 2018).

As a standard step in NLP pipelines, methods like tokenisation, removing stop words, stemming, and assigning term importance (e.g., TF-IDF) are frequently used to structure raw text for classification and to improve model performance (Kaushik and Naithani, 2015).

Tools Commonly Used

Several sentiment analysis tools are available, ranging from open-source libraries to commercial platforms:

VADER: Designed for platforms like social media and brief textual inputs in sentiment evaluation.

TextBlob: Provides a straightforward interface for typical NLP operations, including built-in emotion detection.

Scikit-learn: Delivers a broad selection of machine learning algorithms for classifying sentiment.

BERT-based models (e.g., HuggingFace Transformers): Provide contextualised embeddings for advanced sentiment analysis tasks.

RapidMiner and Weka: GUI-based platforms supporting ML-driven sentiment workflows.

Google Cloud NLP and IBM Watson NLU: Offer sentiment analysis as cloud-based services.

These tools vary in complexity and customisation, allowing researchers to select solutions that best match the scale and scope of their analysis.

2.3 Topic Modelling in Customer Feedback

As customer feedback increasingly takes the form of detailed, unstructured online reviews, the ability to uncover meaningful patterns within this content has become crucial. Topic modelling falls under unsupervised machine learning and helps identify underlying topic patterns in large text datasets by grouping together frequently co-occurring terms. Within the framework of sentiment analysis, topic modelling serves to detect and classify the most relevant elements or themes customers mention in their reviews.

A widely adopted method in this area is Latent Dirichlet Allocation (LDA), which treats each text as a blend of multiple topics and each topic as a probabilistic distribution over words. In the

context of customer reviews, it can reveal recurring themes like product features, pricing, or support experiences. These identified themes give brands meaningful insight into frequently discussed concerns and how sentiment is distributed across them.

Topic modelling also supports aspect-level sentiment analysis, which moves beyond general sentiment tagging by linking opinions to particular product or service attributes. For example, a customer may express satisfaction with a Nike shoe's design but frustration with the delivery experience. Topic modelling allows these aspect-level themes to be detected, making it easier for companies to respond to precise areas of concern or praise.

A major strength of topic modelling is its ability to process large datasets without human supervision, making it ideal for analysing thousands of reviews at once. However, like all models, it has limitations. While LDA has demonstrated strong capabilities in identifying clearly expressed topics from reviews, it often struggles to detect implied themes that are not explicitly mentioned within the text (Rana, Cheah and Letchmunan, 2016). This can be a drawback when customers describe experiences indirectly or use subjective expressions that are not easily grouped by topic.

Despite these limitations, topic modelling remains an essential tool in the sentiment analysis toolkit. When combined with sentiment classification techniques, it allows for a dual-layered interpretation of customer reviews — revealing not only what people are talking about, but also how they feel about those topics.

2.4 NLP in Brand Reputation and Customer Insights

In the digital economy a brand's image is no longer crafted solely through advertising or controlled narratives. Instead, it is increasingly shaped by digital discussions, online reviews, and social media interactions. As a consequence, many companies are leveraging Natural Language Processing (NLP) to interpret large-scale user-generated content and extract meaningful findings that reflect public sentiment and brand perception.

NLP technologies like sentiment analysis, aspect detection, and entity recognition enable organisations to track brand perception across digital channels. These tools provide a systematic way to track opinion trends, identify emerging concerns, and measure the impact of marketing campaigns or product launches. For brands like Nike, which operate globally and engage millions of consumers, NLP makes it possible to capture feedback in real time and respond to reputational risks or opportunities more effectively.

A key benefit of NLP in branding is its ability to associate sentiment with particular product or service elements—for instance, identifying whether concerns relate to product quality, late deliveries, or pricing. This aspect-level granularity enables companies to move beyond surface-level metrics such as star ratings and instead focus on the root causes of customer satisfaction or dissatisfaction.

NLP has been instrumental in advancing the automation of brand monitoring systems. Through the collection and interpretation of online feedback, brands can generate dashboards and reports that provide a continuous pulse on consumer attitudes. This supports more informed decision-making across marketing, product development, and customer service functions.

Sentiment analysis enables organisations to systematically analyse what people think about entities such as products, services, or companies (Liu, 2012). By leveraging these techniques,

brands can move from reactive problem-solving to proactive engagement based on data-driven insights.

2.5 Challenges in Analysing Customer Reviews

Although sentiment analysis is widely regarded as an effective method for capturing public sentiment, analysing customer reviews involves various technical and linguistic hurdles. These challenges often stem from the casual, varied, and frequently unclear characteristics of human communication as seen in digital contexts.

A major difficulty lies in handling contextual ambiguity. Identical terms may convey contrasting emotions depending on how they are used—for example, the term “*light*” may be positive for electronic gadgets but negative for construction materials. Machines often struggle to capture such subtle variations in meaning. Similarly, domain-specific language, such as slang, product-related jargon, or abbreviations, can reduce the effectiveness of general-purpose sentiment models.

Another difficulty is handling mixed sentiment within a single review. A customer may praise one aspect of a product while criticising another, making it difficult to determine the general emotional tone of a review using basic positive or negative labels. Standard classifiers often fail to accommodate such nuanced content, especially when sentiment polarity shifts within the same sentence.

Sarcasm, irony, and implicit sentiment are further complications. Sentences such as “Great service — they forgot my order twice!” can easily confuse rule-based and even some machine learning models. These rhetorical devices are especially common in consumer reviews, where users express frustration or humour through indirect language.

The issue of fake or deceptive reviews is also a growing concern. Reviews may be fabricated to promote or discredit a brand, and these can often appear grammatically correct and emotionally charged, making them difficult to detect using basic NLP techniques. This adds a layer of complexity to ensuring data quality and reliability.

Moreover, sentiment expression varies across cultures, languages, and regions, affecting both the vocabulary and emotional tone of reviews. Even within English-language texts, regional idioms or writing styles can affect how sentiment is conveyed and interpreted.

It has been noted that sarcastic remarks such as “My order was delayed. Very good.” are easily understood by humans but continue to challenge NLP models due to their layered emotional cues (Shivaprasad and Shetty, 2017).

In response to these challenges, ongoing research is focused on enhancing NLP techniques—such as integrating contextual embeddings, sentiment detection at multiple levels, and recognising sarcasm—to strengthen the precision and reliability of sentiment analysis systems.

2.6 Research Gaps and Contribution

Although notable advancements have been achieved in sentiment analysis through both traditional as well as deep learning techniques, several research gaps persist in the current literature. Most existing studies tend to focus heavily on overall sentiment polarity without adequately addressing the underlying causes or specific aspects contributing to those sentiments. This limitation restricts the usefulness of such analyses for actionable business

insights, especially in domains like brand management where understanding the *why* behind customer sentiment is crucial.

Another gap lies in the handling of linguistic complexity. Customer reviews often feature sarcasm, casual language, or blended opinions—challenges that many rule-based and machine learning systems continue to face in terms of accurate interpretation. Even though advanced deep learning architectures like LSTM, GRU, and BERT have demonstrated better performance in recognising context, limitations still exist when dealing with ambiguity, long-distance dependencies, and negation.

Additionally, there is a limited amount of work that combines both sentiment classification and topic modelling in a comprehensive way to extract deeper insights from unstructured text. Most research either focuses on identifying sentiment polarity or on topic extraction, but not on their intersection — such as how certain topics drive specific sentiments over time or across user demographics.

Many existing models also face performance trade-offs between accuracy, interpretability, and computational efficiency. While deep learning delivers high accuracy in various applications, these models typically demand substantial labelled data, extended training durations, and intensive computational power—making them less practical for everyday business scenarios.

It has been observed that despite the improved accuracy of deep learning techniques, many models still struggle with semantic coherence and context interpretation, especially when applied across different domains and writing styles (Islam, Ab Ghani and Ahmed, 2020).

Contribution of This Study

This dissertation contributes to addressing these gaps by:

- Integrating sentiment detection with topic modelling to offer a multi-dimensional view of customer reviews.
- Using Trustpilot data specific to Nike, allowing for brand-specific and aspect-focused insights.
- Applying both VADER for sentiment scoring and LDA for topic extraction, followed by trend analysis across time and regions.
- Presenting practical visualisations and structured outputs that can support brand strategy and customer experience improvement.

This integrated approach enhances interpretability while maintaining computational efficiency, offering a model that could be adapted by practitioners in marketing, branding, and data science roles.

2.7 Summary

This chapter offered a thorough summary of the main academic concepts linked to this research. It opened by outlining sentiment analysis as a key element of natural language processing, emphasizing its function in interpreting public sentiment based on unstructured textual content. A variety of tools and techniques were reviewed, spanning from rule-based methods like VADER and TextBlob to advanced approaches in machine learning and deep learning, such as BERT, which enhance accuracy and contextual awareness.

Topic modelling was then explored as a complementary method for identifying underlying themes in customer feedback. Its integration with sentiment analysis enables a deeper, aspect-based interpretation of consumer attitudes. The chapter also examined how these technologies are applied in brand management, allowing companies to monitor reputation, detect emerging issues, and improve customer engagement through data-driven insights.

Despite the progress in the field, several challenges remain — including sarcasm detection, domain-specific language, and the handling of mixed or implicit sentiment. The literature further reveals that many studies focus on sentiment or topics in isolation, with limited attention paid to their intersection or temporal and regional variations. These gaps provide an opportunity for this research to contribute a more integrated and applied approach to analysing customer sentiment.

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the methodological approach used to carry out a comprehensive sentiment and topic analysis of Nike customer reviews sourced from Trustpilot. The study adopts a mixed-methods strategy within a data science framework, leveraging natural language processing (NLP) and unsupervised machine learning to derive useful insights from unstructured textual input...

The workflow includes five main stages: acquiring data, preparing it, performing sentiment classification, modelling topics, and visualisation. Each step is described in detail, including techniques, models, and tools. Python was chosen primarily for its broad ecosystem of data analysis libraries. All processes were executed on Google Colab, which enabled scalable processing via cloud infrastructure.

Sentiment classification was carried out using the VADER tool, and topic extraction employed Latent Dirichlet Allocation (LDA). These tools were selected based on their effectiveness in analysing short, content-rich review text. Various libraries were used for visualisation to help interpret the findings clearly.

The chapter also covers ethical protocols and discusses the practical challenges encountered during the course of the study.

3.2 Data Collection

This study used a dataset comprising reviews of Nike products, gathered from the Trustpilot platform. Trustpilot is an openly accessible review portal that features extensive consumer feedback across multiple industries. It was chosen as the main data source due to its rich repository of genuine, text-based reviews, ideal for sentiment and topic analysis.

Data was obtained using a publicly accessible web-scraping utility, which was employed to extract reviews directly from Trustpilot's website. This tool enabled automated retrieval of Nike customer reviews by scraping the HTML content of relevant review pages from Google search results and Trustpilot. The extracted data included key fields such as the full text of each review, the review date, the star rating (where available), and the reviewer's location if specified. This

method allowed for efficient collection of a large volume of text-rich, user-generated content without the need for custom Python scripts.

In total, approximately 5,000 English-language reviews were extracted, covering a time span of several months to ensure sufficient diversity in customer experiences. The reviews were saved as CSV files in a structured format to support subsequent data analysis in Google Colab. Any reviews written in non-English languages were excluded to maintain consistency in language-based NLP processing.

As Trustpilot only provides anonymised reviewer data, this approach adhered to ethical research protocols and respected data protection standards. All data was used exclusively for academic and research-related purposes.

3.3 Data Preprocessing

To begin the analysis, several preprocessing procedures were applied to the raw reviews collected from Trustpilot to clean and structure the text for use in sentiment detection and topic extraction. Text preprocessing plays a vital role in NLP workflows, as raw input often includes inconsistencies, irrelevant data, or noise that can hinder model performance.

3.3.1 Initial Cleaning

The dataset was initially imported into a pandas DataFrame using Google Colab, followed by the steps below:

- **Removal of null values:** Rows containing missing or empty review texts were removed to ensure all data points were valid for analysis.
- **Removal of duplicates:** Duplicate reviews were dropped to avoid overrepresentation of particular customer opinions.
- **Language filtering:** Only English-language reviews were retained for consistency in text processing and compatibility with tools such as VADER and LDA.

3.3.2 Text Normalisation

Next, the text of each review underwent standard NLP normalisation using Python libraries such as re, nltk, and string, following widely adopted methods outlined in standard NLP resources (Bird, Klein and Loper, 2009).

- **Lowercasing:** All text was transformed to lowercase to eliminate case-based redundancy (e.g., treating “Least” and “least” as equivalent).
- **Punctuation removal:** Punctuation symbols were stripped out to streamline tokenisation.
- **Stopword removal:** Frequently used English words (e.g., “the”, “as”, “and”) were filtered out using NLTK’s built-in stopwords list.
- **Tokenisation:** Each review was divided into individual tokens or words.
- **Lemmatisation:** Words were normalised to their root forms via the WordNetLemmatizer, helping unify different forms of the same word (e.g., “walking”, “walks” to “walk”).

Text preprocessing is known to significantly enhance sentiment analysis performance by removing noisy elements and making the input more structured (Chandrasekaran, Nguyen and Hemanth, 2021).

3.3.3 Review Filtering

To ensure textual quality and prevent model bias, extremely short reviews (e.g., fewer than three words) were discarded, as they typically lacked meaningful content. Reviews that contained only symbols or irrelevant characters were also excluded.

The resulting cleaned and lemmatised dataset was stored in a new column and used for both sentiment classification and topic modelling. This rigorous preprocessing step enhanced the performance and reliability of the downstream models by reducing noise and increasing linguistic consistency across the corpus.

3.4 Sentiment Analysis

Sentiment analysis was conducted to categorise each review as positive, negative, or neutral, depending on its emotional content. This step aimed to quantify customer attitudes toward Nike and observe how sentiments varied across different customer experiences. A lexicon-based approach was selected for its effectiveness in processing short, informal, and opinion-heavy texts commonly found in customer reviews.

The method used was VADER (Valence Aware Dictionary and sEntiment Reasoner), which is a lexicon-based tool that applies a predefined set of lexical indicators to generate sentiment scores. VADER is especially effective for analysing social media-like content, as it can process features like capitalisation, punctuation, intensity modifiers, and even emoticons. Each review is given a compound sentiment score, with values ranging between -1 (strongly negative) and +1 (strongly positive). Sentiment is determined using these thresholds:

- Positive sentiment: compound score ≥ 0.05
- Neutral sentiment: $-0.05 < \text{score} < 0.05$
- Negative sentiment: compound score ≤ -0.05

This analysis was carried out using Python's vaderSentiment library in Google Colab. After preprocessing, each review was passed through the analyser to generate a compound score and assign it a sentiment label. These sentiment labels were stored in a separate column and later used for trend analysis and cross-comparison with extracted topics.

Lexicon-based models like VADER require no training data, making them suitable for projects with limited labelled datasets. They also offer transparency in how sentiment scores are derived, unlike many black-box machine learning approaches. Sentiment analysis enables researchers to computationally classify emotional tones in public communication, offering scalable insight into consumer opinion without manual intervention (Praveen and Ittamalla, 2021).

The sentiment-labelled dataset served as a foundation for subsequent phases of this research, including visualisation, topic correlation, and temporal sentiment tracking.

3.5 Topic Modelling

To complement the sentiment analysis and uncover deeper themes in customer feedback, this research implemented Latent Dirichlet Allocation (LDA) for topic modelling. LDA is an

unsupervised learning method that identifies patterns of co-occurring words, helping interpret unstructured content effectively.

The modeling process was executed using the Gensim package in Python. Preprocessed and lemmatised reviews were vectorised with CountVectorizer to create a document-term matrix suitable for modelling. Several topic counts were tested iteratively, with the final selection based on achieving a balance between coherence score and interpretability.

Each review was assigned a probability distribution across the topics, and the dominant topic was extracted for further analysis. Key terms from each topic were extracted to support interpretation and assist with manual labelling.

LDA operates under the assumption that each document contains a blend of topics, where each topic consists of a set of words with varying probabilities. This statistical method supports the discovery of overlapping ideas within individual reviews. It offers a scalable solution for analysing document-level structure in text data, and is particularly effective in identifying patterns within large datasets (Blei, Ng and Jordan, 2003).

The topic labels were later cross-referenced with sentiment scores to explore how customer attitudes differed across themes, as well as how specific issues influenced negative or positive sentiment. This dual-layered insight enhances the practical relevance of the findings for brand managers and marketing teams.

3.6 Visualisation and Tools

To support analysis and improve the interpretability of results, a variety of visualisation techniques were used throughout the project. Visualisations played a critical role in highlighting sentiment distributions, topic relevance, keyword frequency, and temporal trends within the customer review data.

The full analysis was conducted using Google Colab, a cloud-based platform that integrates seamlessly with Python tools and allows for live collaboration. Colab was selected for its accessibility, user-friendliness, and capability to manage large-scale data processing without depending on local hardware resources.

3.6.1 Tools and Libraries

Several Python libraries were employed to carry out the data analysis and generate visual outputs:

- **Pandas:** Used for reading, cleaning, and manipulating the dataset in tabular form.
- **Matplotlib** and **Seaborn:** Used to produce visual plots such as bar graphs, pie charts, histograms, and time-series visualisations that illustrate sentiment breakdowns and changes across time
- **WordCloud:** Used to visualise the most frequent words in the corpus and within specific topics.

- **Gensim** and **pyLDAvis**: Gensim was used to build the LDA model, while pyLDAvis provided an interactive interface for exploring topic structures, keyword weights, and document-topic distributions.
- **NLTK** and **spaCy**: These libraries were used to carry out NLP operations including tokenisation, stopword filtering, and lemmatisation.
- **VADER** (via vaderSentiment): Used for calculating sentiment scores for each review.

3.6.2 Visualisation Examples

Visualisations were incorporated throughout the analysis to enhance interpretability and communicate findings effectively. Key outputs included sentiment distribution charts, time-series plots, word clouds, and topic maps.

Sentiment charts and time-series visualisations were applied to show the distribution across positive, negative, and neutral sentiment categories, and to observe sentiment trends over time. Word clouds helped highlight commonly occurring terms in the review corpus, while topic visualisations provided insights into theme prevalence and keyword relevance.

Although the interactive pyLDAvis tool could not be used due to technical constraints, alternative visual summaries were employed to validate and present the topic model. These visual tools supported better comprehension of sentiment-topic patterns for both expert users and general audiences.

3.7 Ethical Considerations

This research was carried out with a strong focus on ethical standards, especially concerning data protection, informed consent, and responsible usage of publicly available content. All data was collected from Trustpilot, a publicly accessible platform where reviews are voluntarily posted by users without any access restrictions or login requirements.

No sensitive personal information (PII) was gathered, stored, or handled during data collection. The dataset consisted exclusively of review text, review date, and general metadata such as the star rating and country (where specified). Sensitive identifiers such as usernames, email addresses, or IP data were neither accessed nor extracted, ensuring full compliance with ethical data handling principles.

All analysis was conducted for academic use only, and the dataset was accessed exclusively via a secured environment on Google Colab. Only the research team had file access, and no content was redistributed for profit.

Although formal ethical clearance was not necessary due to the anonymous, publicly available nature of the dataset, the research followed principles of data minimisation, context sensitivity, and harm prevention. This aligns with the ethics framework for internet-based research outlined by the Association of Internet Researchers, which emphasises context-specific, case-driven decision-making (Markham and Buchanan, 2012).

3.8 Limitations

The data used in this study was solely retrieved from Trustpilot, a single review website. While Trustpilot offers valuable consumer insights, it may not capture sentiment accurately across

other platforms like Twitter, Reddit, or those tied to specific brands. As a result, findings could reflect platform-specific tendencies, which may reduce their relevance for broader populations.

3.8.1 Platform Dependency

All data was collected exclusively from Trustpilot, a single review platform. While Trustpilot provides high-quality consumer feedback, it may not fully represent the sentiments expressed on other platforms such as Twitter, Reddit, or brand-specific forums. As a result, the insights derived may reflect platform-specific biases, limiting their applicability across wider consumer segments.

3.8.2 Language Constraints

Only reviews written in English were included in the analysis. This approach excludes non-English-speaking users, potentially omitting culturally diverse perspectives. Sentiments and expressions can vary significantly across languages, which limits the cross-cultural relevance of the findings.

3.8.3 Lexicon-Based Sentiment Classification

VADER, the sentiment detection method applied here, is a rule-based system tailored for short-form and informal text. While it works well for rapid classification, it lacks the nuanced understanding and adaptability of more sophisticated deep learning tools. This may limit its effectiveness when interpreting sarcasm, subtle emotion, or mixed tones within a single review.

3.8.4 Interpretability of LDA

Latent Dirichlet Allocation (LDA) was used for topic extraction because of its balance between simplicity and robustness. However, LDA treats topics as distinct and fixed, which can be a mismatch for the nuanced and overlapping nature of consumer feedback. Additionally, interpreting the results often involves a level of subjectivity, potentially introducing researcher bias.

3.8.5 Temporal and Sentiment Generalisation

Although some time-based analysis was performed, the distribution of reviews over time was uneven, potentially skewing temporal sentiment trends. Additionally, assigning a single sentiment label to each review may not reflect the full emotional complexity, especially when contrasting opinions exist within the same review.

Despite these limitations, the methods chosen were appropriate for the research objectives and provided valuable insights into customer perceptions of Nike. The study maintains transparency and methodological consistency while laying the groundwork for future improvements and deeper analysis.

3.9 Chapter Summary

This chapter summarised the methodology used to explore customer sentiment and extract key themes from Nike product reviews sourced from Trustpilot. It began by explaining how data was gathered, including scraping openly available reviews and limiting the analysis to English-language content to ensure consistency.

The preprocessing section described steps taken to prepare the textual data, including text segmentation, root word extraction, and stopword elimination to enable effective NLP

processing. Sentiment was then analysed using VADER, a lexicon-driven tool designed for brief and casual text. Each review was assigned a sentiment category—positive, neutral, or negative—based on a calculated compound scores.

To conduct topic analysis, the study applied the Latent Dirichlet Allocation (LDA) technique to identify key themes within the dataset. This helped identify dominant topics discussed by customers and allowed for the integration of sentiment trends across thematic areas. Visualisations were used extensively to represent sentiment distributions, topic clusters, and keyword frequencies, enhancing the interpretability of results.

The chapter also addressed ethical considerations, highlighting the use of publicly accessible, anonymised data in compliance with data protection principles. Lastly, it acknowledged key limitations, including platform constraints, language filtering, and the interpretability of models used.

Together, these methodological components provided a rigorous and replicable framework for analysing large-scale customer feedback in a brand-specific context.

Chapter 4: Analysis and Results

4.1 Introduction

This section presents the outcomes of sentiment analysis and topic modelling conducted on Nike customer reviews obtained from the Trustpilot website. The analysis aims to examine consumer opinions, uncover common themes in feedback, and assess how sentiment varies across topics and time periods.

The findings are structured to reflect the methodological stages described in the previous chapter. The results begin with a summary of the dataset and proceed to the findings of the VADER sentiment classification, including how sentiment labels are distributed, examples of review content, and sentiment patterns over time. Bar graphs and word cloud visualisations are used to highlight frequent terms and emotional expressions in the data.

The section then details the findings from topic modelling using LDA, including keywords, illustrative review samples, and topic visualisations. Lastly, the connection between topics and sentiment categories is explored to understand what themes influence positive or negative sentiment.

This results chapter is visual in nature, relying on charts, tables, and diagrams to support interpretation and enable a clearer understanding of patterns in the data.

4.2 Dataset Overview

The dataset used in this analysis comprises Nike product reviews collected from the Trustpilot website. The raw data included more than 5,000 entries, each containing customer-submitted textual feedback, alongside metadata such as the date of the review and, where available, the reviewer's region. After conducting cleaning and preprocessing steps, a final set of 4,765 reviews was selected for analysis.

These entries span a timeframe of over 15 years, from December 2009 to April 2025, offering a strong temporal dataset to explore how customer sentiment and thematic trends evolved.

The reviews averaged 45 words each, with extremely short responses removed to ensure meaningful content remained.

During preprocessing, reviews were converted to lowercase, punctuation and stopwords were eliminated, and lemmatisation was applied. Reviews written in other languages, duplicated entries, and submissions with fewer than three words were excluded to maintain consistency and data integrity.

Unlike some customer review datasets, this dataset did not include numerical ratings such as star scores. The analysis was therefore based solely on the text content of reviews and associated metadata (date and region). This provided a focused lens for examining sentiment and topic patterns derived from natural language alone.

The table below summarises key attributes of the dataset following the cleaning process:

Total reviews scraped	~5000
Reviews after cleaning	4765
Time period covered	Dec 2009 – Apr 2025
Language filtered	English only
Average review length	45 words
Min. review length (after filter)	3 words
Star rating available	No
Location metadata included	Partial (region only)

Table 1: Cleaned Dataset Summary

This refined dataset served as the foundation for both sentiment analysis and topic modelling procedures in the research. The following section outlines how sentiment was analysed, including the breakdown of responses into positive, neutral, and negative categories.

4.3 Sentiment Analysis Results

This section reports the output of sentiment analysis conducted using the VADER (Valence Aware Dictionary and sEntiment Reasoner) tool. VADER is a rule-based model developed for analysing brief and informal text, such as customer reviews and comments on social platforms. It generates a compound score for each review, ranging from -1 (strongly negative) to +1 (strongly positive), which is then used to classify sentiment into the following three groups:

- Positive: compound score ≥ 0.05
- Neutral: $-0.05 < \text{score} < 0.05$
- Negative: compound score ≤ -0.05

The model was applied to the cleaned dataset of 4,765 Nike customer reviews. Each review was passed through the analyser, and the resulting sentiment label was recorded for further analysis.

4.3.1 Sentiment Distribution

The sentiment categories identified throughout the dataset are illustrated in the accompanying figure. The outcome reveals a polarised trend, with positive and negative reviews occurring in nearly equal proportions, and a considerably lower share of neutral responses.

- Positive: approximately 2,200 reviews (46.2%)

- Neutral: approximately 350 reviews (7.3%)
- Negative: approximately 2,215 reviews (46.5%)

This pattern suggests that while many customers express satisfaction with Nike products and services, a similarly large portion voice dissatisfaction, reflecting a divided sentiment landscape.

The chart below depicts how many reviews fall into each sentiment group, as classified through the VADER sentiment model.

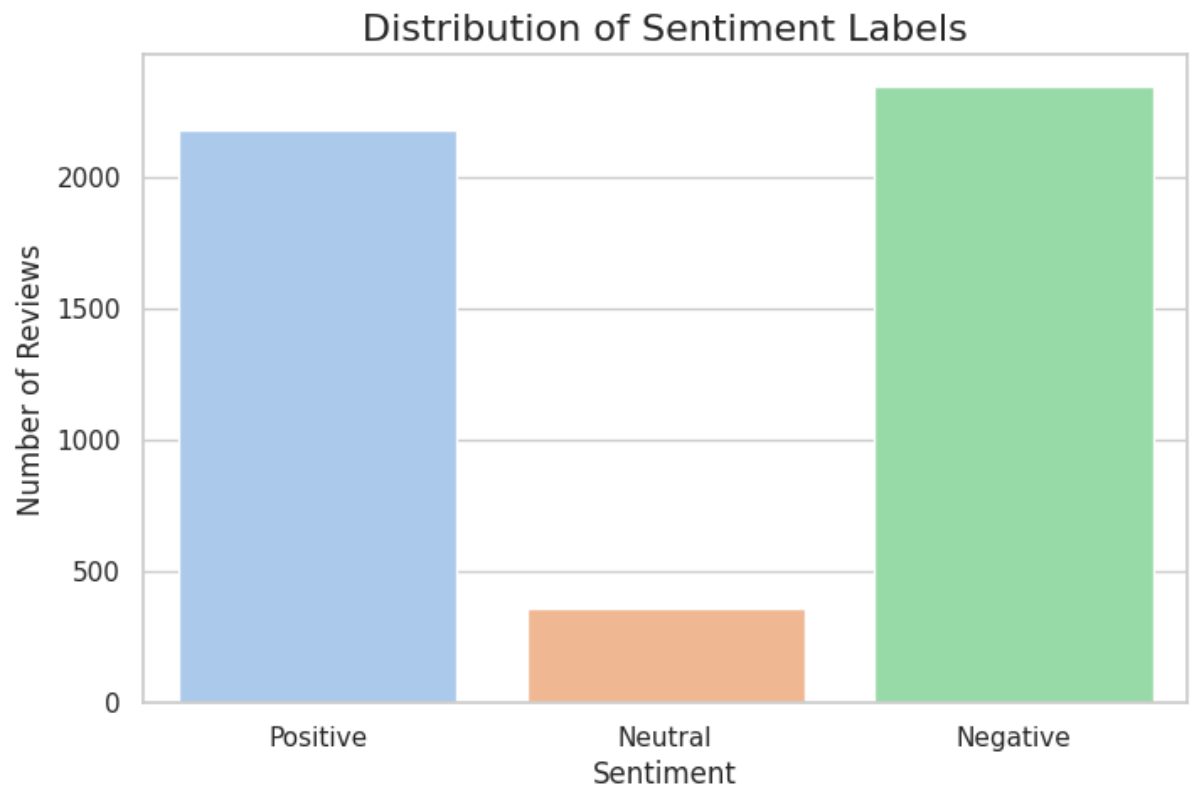


Figure 1: Distribution of Sentiment Categories

4.3.2 Sample Reviews by Sentiment

To illustrate how sentiment labels were assigned using VADER, Table 2 presents selected review excerpts along with their corresponding sentiment classification and dominant topic assignment. These examples reflect the variety of experiences expressed by Nike customers and demonstrate how the VADER algorithm evaluates emotional tone.

No.	Review	Cleaned Review	Sentiment Label	Dominant Topic
1	Terrible customer service I was told that I wa...	terrible customer service told crazy problem w...	Negative	5
2	they have the best shoes. the dunks are comfor...	best shoe dunk comfortable ordered nike cam...	Positive	3

3	Shitty company with paper releases that only	shitty company paper release get sold reseller	Negative	4
4	Mislead information by one assistant on live c...	mislead information one assistant live chat se...	Negative	2
5	Always the same size with the same model for years including costumize . Suddenly they fall a size 1/2 smaller then normal.	always size model year including costumize suddenly fall size smaller normal	Neutral	3
6	The shoes are amazing comfortable and affordable	shoe amazing comfortable affordable	Positive	3
7	It was doable, I guess	doable guess	Neutral	4
8	Very nice lady called Anita got my refund sorted in 5 minutes and the money is on the way please give her a raise	nice lady called anita got refund sorted minute money way please give raise	Positive	2
9	If i could rate them zero I would. I will neve...	could rate zero would never buy another nike p...	Positive	1

Table 2: Example Reviews with Sentiment Labels

*Note: The final review was misclassified as positive by VADER, despite containing clear negative sentiment. This reflects a known limitation of rule-based sentiment analysis tools when handling sarcasm or negations.

These examples show that VADER generally performs well when reviews contain explicit emotional language (e.g., “terrible”, “great”). However, its performance can be limited when dealing with implicit cues, mixed sentiment, or complex phrasing. As seen in the misclassification above, VADER may occasionally overlook sentiment direction if the sentence structure contradicts the typical word polarity.

While sentiment labels provide a high-level overview of customer opinion, they do not explain *what* customers are talking about. For a more detailed understanding of the underlying issues driving satisfaction or dissatisfaction, the next section applies topic modelling to extract dominant themes from the review corpus.

4.4 Word Frequency and Word Clouds

Gaining insight into the most commonly used words in customer feedback offers a clearer understanding of Nike customers’ key concerns, compliments, and experiences. Word frequency analysis was applied to the cleaned review dataset to identify common themes before deeper topic modelling. This chapter presents a series of word clouds and bar plots to visualise and quantify term usage, both across the entire dataset and within individual sentiment categories.

Before conducting this analysis, all reviews were preprocessed. This included lowercasing the text, stripping punctuation and stopwords, and performing lemmatisation to unify different forms of the same word (e.g., “run” and “running”). These steps ensured that the visualisations accurately reflected meaningful content rather than grammatical noise.

4.4.1 Overall Word Frequency

Two types of visualisations were generated to assess word frequency across the entire dataset, irrespective of sentiment:

1. **Word clouds** to intuitively highlight commonly used terms.
2. **Bar charts** were employed to measure and display the 20 highest-frequency terms.

Word clouds are widely used in text analytics for their ability to provide a quick, visual overview of dominant terms, especially in early exploratory analysis (Heimerl et al., 2014).

This word cloud displays unprocessed terms from the raw dataset. Common stopwords such as “the”, “I”, “to”, “my”, and punctuation marks dominate the display, obscuring more relevant content.

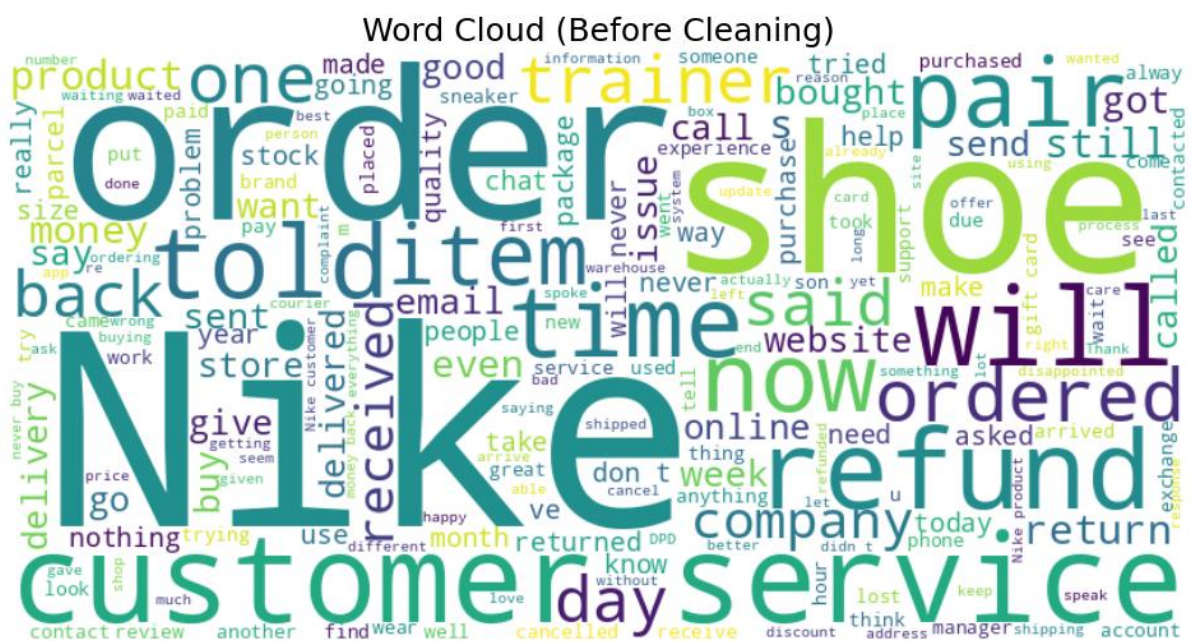


Figure 2: Word Cloud Before Text Cleaning

Following the preprocessing stage, the refined word cloud showcases key terms related to content such as “nike”, “order”, “customer”, “refund”, “shoe”, and “service”. These words suggest a strong focus on ordering experience, product satisfaction, and customer service.

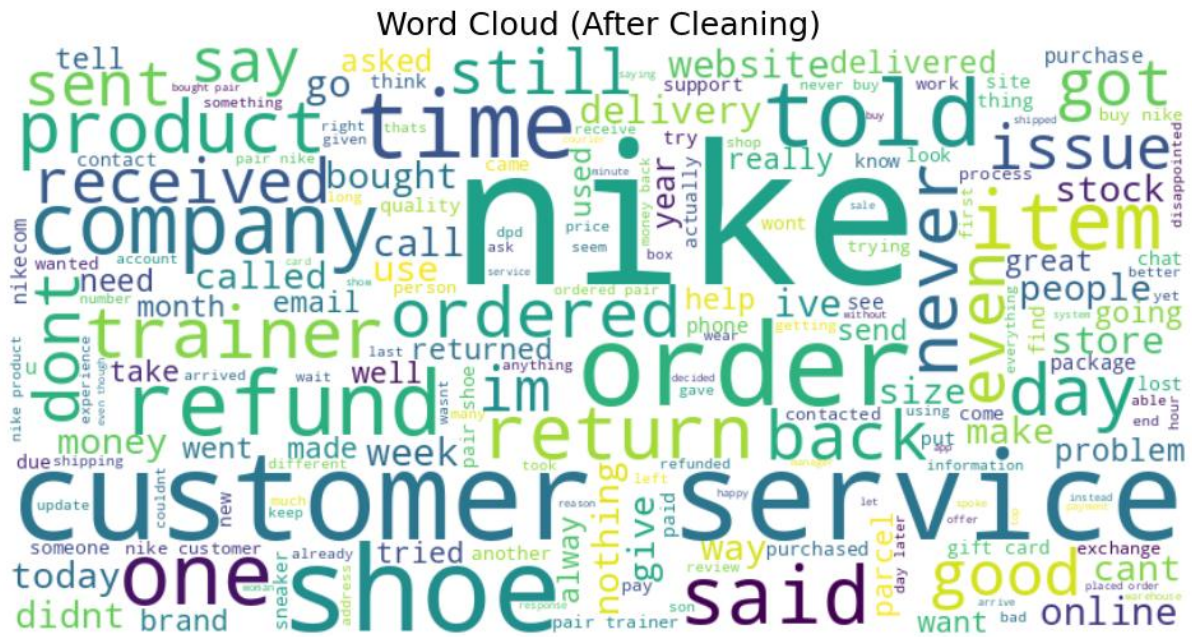


Figure 3: Word Cloud After Text Cleaning

This bar plot reinforces the presence of stopwords in raw data, demonstrating the need for thorough preprocessing. Words like “the”, “I”, “and”, “to”, and punctuation still appear frequently.

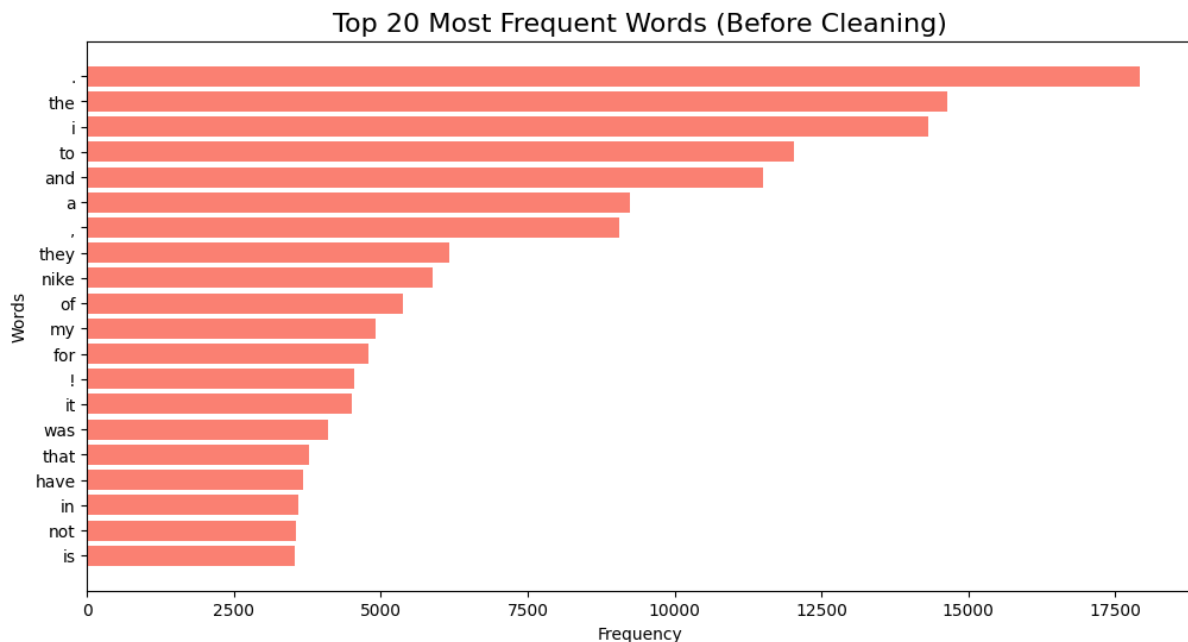


Figure 4: Most Common Words Prior Cleaning

In the cleaned data, “nike” is the most common term, followed by “order”, “customer”, “shoe”, and “service”. These terms align with the topics reflected in customer experiences, such as

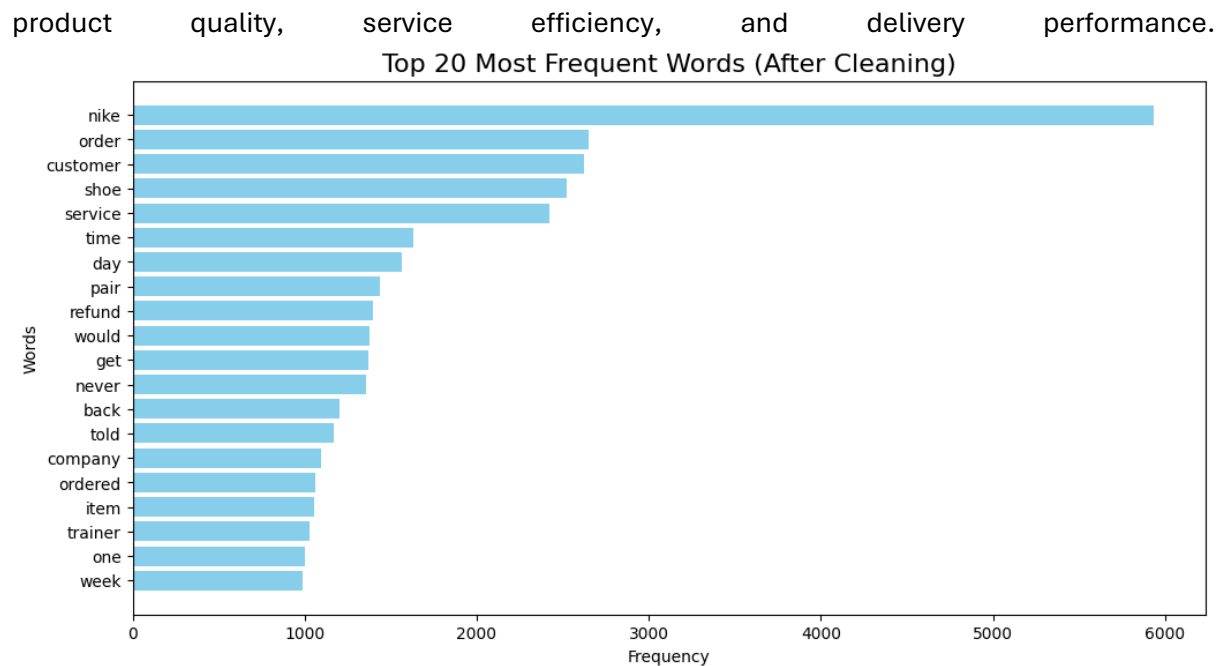


Figure 5: Most Common Words After Cleaning

Together, these visualisations confirm that customer conversations revolve around Nike’s products, ordering process, and customer service. These insights set the stage for more structured thematic analysis in the form of topic modelling.

4.4.2 Word Clouds by Sentiment

This visual representation illustrates the frequently occurring terms associated with both positive and negative sentiments in customer reviews, enabling a comparative analysis of sentiment-driven language.

The left half of the figure, representing positive reviews, is dominated by words such as “nike”, “shoe”, “order”, “comfortable”, “great”, “delivery”, and “service”. These terms suggest that positive sentiment is largely driven by product satisfaction, good fit, timely delivery, and positive store experiences. Words like “love”, “perfect”, and “excellent” indicate strong approval of Nike’s offerings.

The right half of the figure, showing negative reviews, highlights terms such as “refund”, “return”, “problem”, “delay”, “told”, “called”, and “customer service”. These reflect challenges encountered after purchase, particularly concerning delivery, order processing, and customer service interactions.

This contrast in language between sentiment categories provides insight into the underlying causes of customer satisfaction and dissatisfaction. While positive sentiment tends to focus on product quality and seamless delivery, negative sentiment is heavily tied to service recovery failures, such as delayed responses, return complications, or communication gaps. These findings align with the results from earlier sentiment analysis and set the foundation for the more structured theme extraction carried out in the next section through topic modelling.

Based on this analysis, a 5-topic model was selected. This choice offered an optimal trade-off between model performance and interpretability. Manual review of the resulting topics confirmed that the 5-topic model generated distinct, coherent, and thematically meaningful clusters, such as delivery delays, refund issues, customer support experiences, and product satisfaction. In contrast, models with higher topic counts often fragmented meaningful themes or produced redundant clusters.

This approach aligns with best practices in topic modelling, where quantitative evaluation (e.g., perplexity or coherence) is paired with qualitative review to select the most insightful topic structure.

4.5.1 Overview of Topics

The table below presents the five final topics, each labelled based on its top keywords and dominant narrative.

Topic Label	Description
Customer Support Issues	Complaints about unresponsive, delayed, or poor support interactions
Product Quality & Purchase	Experiences with product fit, comfort, appearance, or value
Returns & Refunds	Issues with refund timelines, return approvals, or policy confusion
Shipping & Delivery	Delays, fast deliveries, or lost items in transit
Website/App Experience	Frustration with website navigation, payment, or order errors

Table 3:Final Topic Labels and Descriptions

These labels were assigned based on frequent keywords and dominant themes emerging in each topic cluster. The resulting classification enabled further cross-analysis with sentiment scores to understand which themes were more polarizing or problematic.

4.5.2 Topic Visualisation

To evaluate how clearly defined and distinct the topics were, an attempt was made to use pyLDAvis for interactive visualisation of the LDA-generated topics. This tool is commonly used to plot topic distributions in two-dimensional space, showing how well topics are separated and which ones may share overlapping vocabulary.

However, due to dependency conflicts and compatibility limitations within the current Python environment, the pyLDAvis tool could not be successfully executed or included in this document. Despite this, topic coherence was thoroughly validated by:

- Examining the most representative keywords for each topic,
- Comparing these keywords to representative customer reviews, and
- Assigning clear, interpretable labels to each topic cluster.

The structure of the five-topic model was therefore considered valid based on qualitative review and practical alignment with the known dimensions of customer experience (e.g., delivery, refunds, customer service).

The next section explores how sentiment is distributed across these five topics to determine which areas drive positive, negative, or neutral customer opinions.

4.6 Sentiment–Topic Relationship

Although topic modelling was utilised to uncover key discussion areas in customer feedback, integrating sentiment classification alongside these topics provided deeper insights into how customers feel about each aspect of their experience. This joint analysis revealed not only what customers were talking about, but also how positively or negatively they felt about those topics.

Each review was labelled with a sentiment category (positive, negative, or neutral) using the VADER sentiment analysis algorithm. Simultaneously, a dominant topic was assigned based on the five-topic LDA model. This dual approach enabled a detailed comparison of sentiment trends across topics and helped assess average sentiment values, offering a clearer understanding of customer satisfaction and dissatisfaction across various areas of Nike’s services and products.

4.6.1 Sentiment Summary by Topic

The table below provides a breakdown of the review counts assigned to each sentiment category—positive, negative, and neutral—across the identified topics.

Topic Label	Positive	Negative	Neutral
Customer Support Issues	380	572	78
Product Quality & Purchase	854	616	124
Returns & Refunds	142	313	33
Shipping & Delivery	577	633	82
Website/App Experience	226	214	43

Table 4: Sentiment Breakdown by Topic

From this table, several patterns are evident:

- **Product Quality & Purchase** had the highest number of positive reviews (854), suggesting strong customer satisfaction with Nike’s product design and value.
- **Shipping & Delivery** and **Customer Support Issues** both had higher negative than positive review counts, indicating recurring problems in logistics and service interactions.
- **Returns & Refunds** was particularly skewed toward negative sentiment, which reflects frustration around refund processes or delays.
- **Website/App Experience** showed relatively balanced sentiment, with both positive and negative feedback suggesting a mixed user experience.

4.6.2 Sentiment Distribution by Topic

The bar chart illustrates the number of reviews associated with each topic, grouped by sentiment category: positive, negative, and neutral.

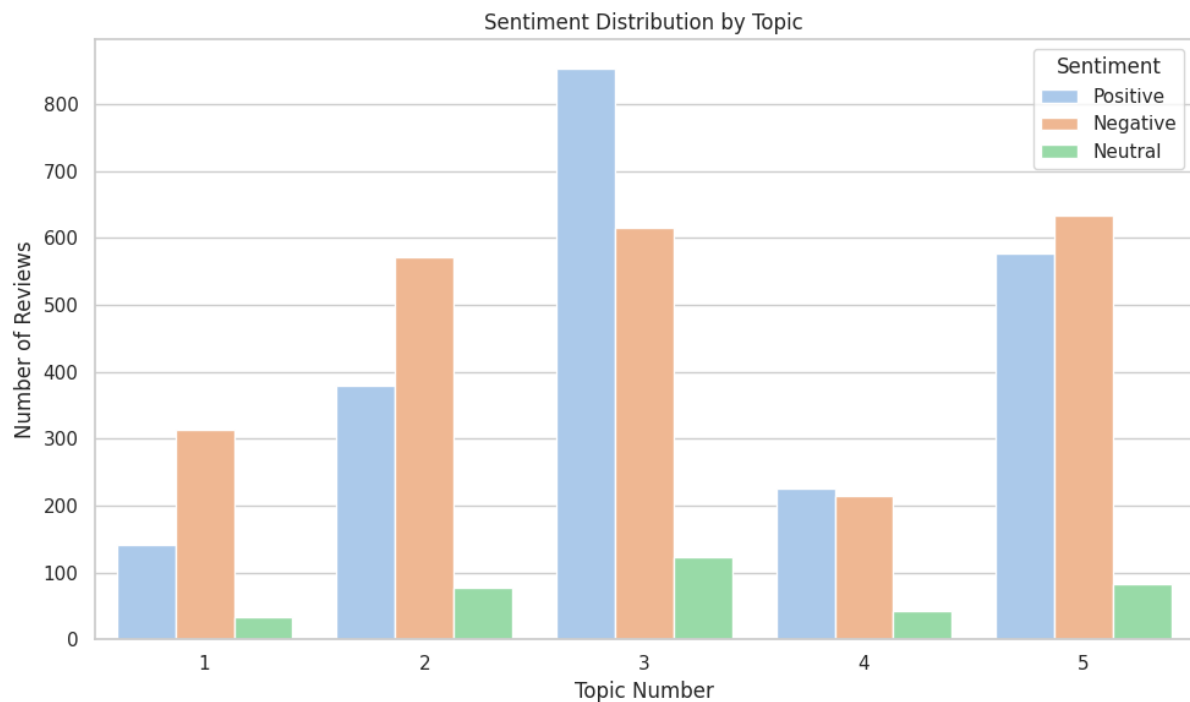


Figure 8: Sentiment Distribution by Topic

Key observations:

- Product Quality & Purchase has the highest number of positive reviews.
- Customer Support Issues and Returns & Refunds are heavily skewed toward negative sentiment.
- Website/App Experience and Shipping & Delivery show a relatively balanced sentiment profile, indicating mixed experiences.

4.6.3 Sentiment Intensity Heatmap by Topic

This section presents a heatmap depicting the relative distribution of sentiment categories—positive, neutral, and negative—across each topic. The values are row-normalised, meaning each row adds up to 1.0, which allows for a direct visual comparison of sentiment concentration within a given topic.

As shown in the figure, the topic “Returns & Refunds” exhibits the highest proportion of negative sentiment, followed closely by “Customer Support Issues,” both exceeding 50% in negative reviews. On the other hand, “Product Quality & Purchase” has the strongest positive sentiment, with over half of the reviews classified as positive. “Shipping & Delivery” and “Website/App Experience” present more balanced sentiment distributions, indicating mixed user experiences. This visualisation offers a complementary perspective to the categorical summaries provided earlier and reinforces the interpretation of sentiment polarity by topic.

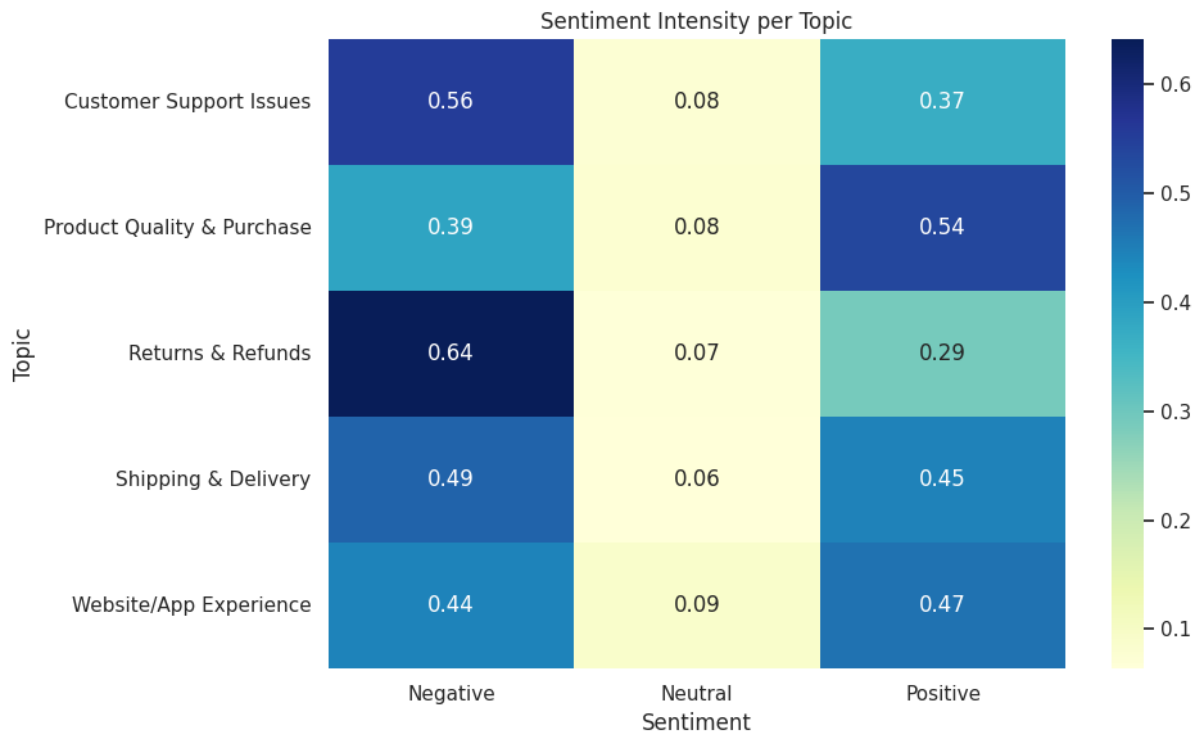


Figure 9:Sentiment Intensity Heatmap by Topic

4.6.4 Average Sentiment Score by Topic

To capture not just the quantity but also the strength of sentiment, the average sentiment score per topic was calculated using VADER's compound score.

This chart plots the mean sentiment score of all reviews within each topic group.

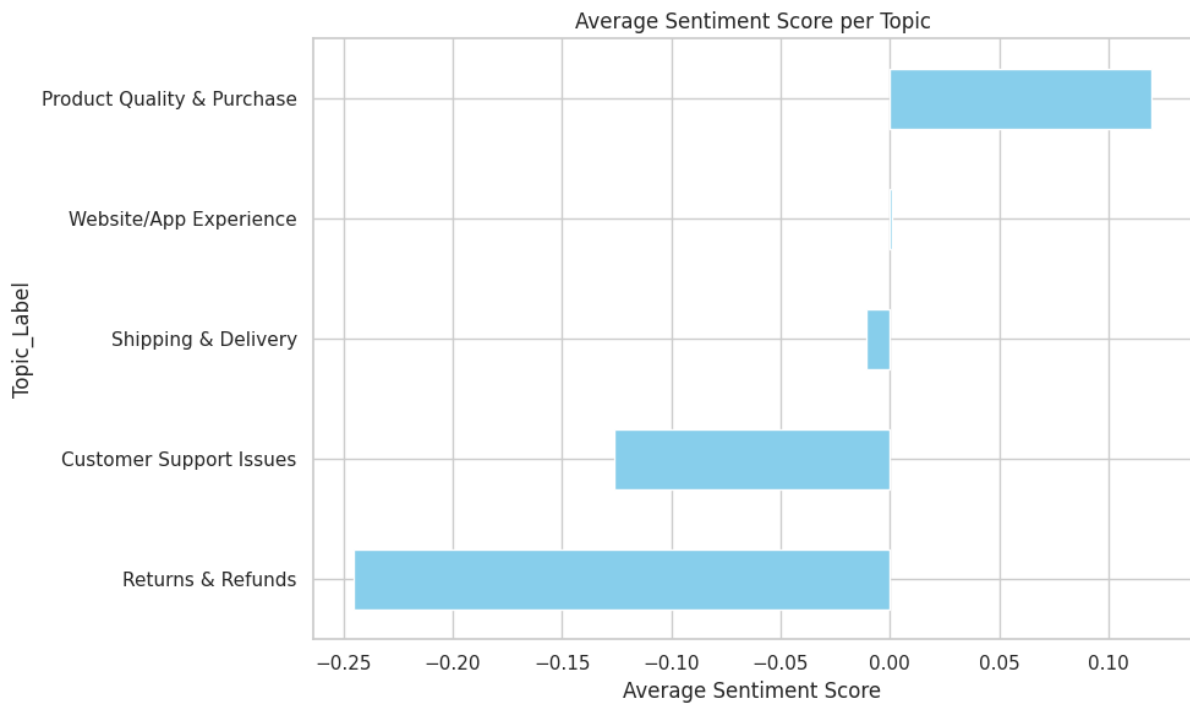


Figure 10:Average Sentiment Score per Topic

Key insights:

- Product Quality & Purchase had the highest average score, reaffirming it as Nike's strongest area.
- Returns & Refunds exhibited the lowest average sentiment, reflecting deeper dissatisfaction.
- Customer Support Issues also showed a negative average sentiment, suggesting recurring service-related frustrations.

4.7 Sentiment Variation by Region and Time

4.7.1 Sentiment by Region

Understanding how sentiment varies across different countries or regions can reveal localised brand perception issues or areas of excellence. Analysing regional sentiment adds geographical context to the overall emotional tone expressed in customer reviews.

The figure showcases the ten regions with the highest and lowest sentiment ratings, calculated from the mean sentiment values derived from customer feedback. The analysis highlights significant geographical variation in how Nike is perceived by its customers.

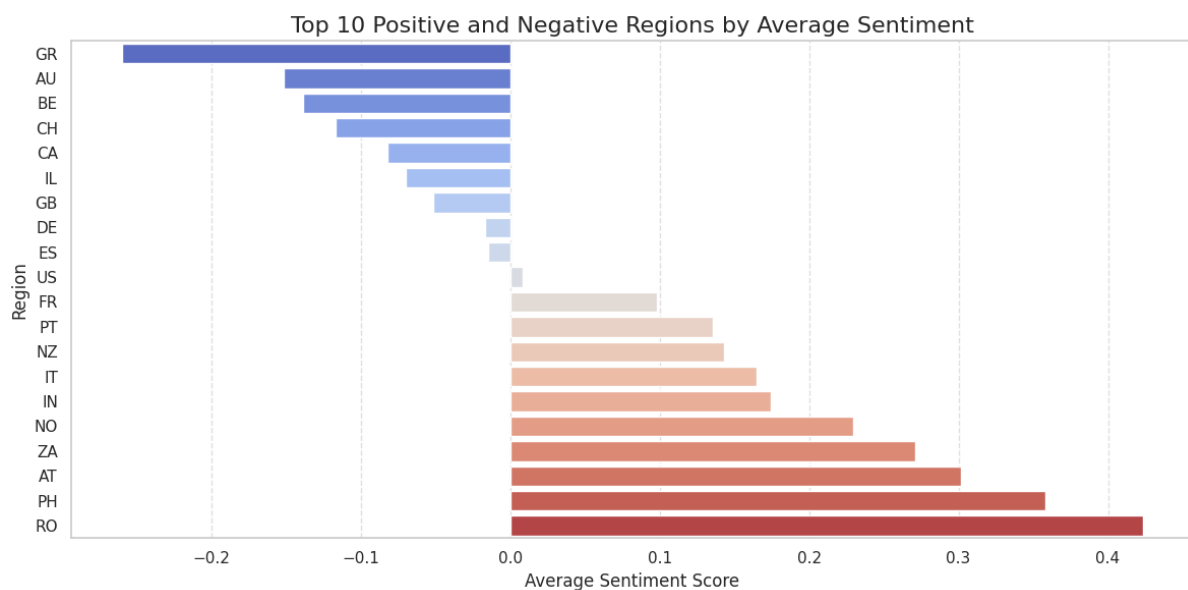


Figure 11: Top 10 Most Positive and Negative Regions by Average Sentiment Score

Figure shows that countries such as Romania (RO), Philippines (PH), and Austria (AT) had the highest average sentiment scores, reflecting strong positivity in their customer feedback. Conversely, countries like Greece (GR), Australia (AU), and Belgium (BE) recorded the lowest average sentiment, suggesting a higher proportion of negative reviews in those regions.

Variations in regional sentiment could stem from multiple influences, such as delivery dependability, efficiency in handling returns, responsiveness of local support teams, and differing cultural norms regarding e-commerce. Despite Nike being a global brand, the findings

indicate that customer sentiment is not uniform across regions and that operational inconsistencies may influence how customers perceive the brand locally.

4.7.2 Sentiment Trend Over Time

To complement the regional analysis, this section explores sentiment variation across time. The figure demonstrates a clear yearly decline in average sentiment scores between 2008 and 2025. While early reviews reflect strong positivity, sentiment has declined over the past decade, particularly post-2016. This shift may be attributed to evolving customer expectations, heightened scrutiny on service quality, or operational challenges. Tracking this trend helps highlight when brand perception began to erode, offering insight into possible inflection points that align with broader strategic or market changes.

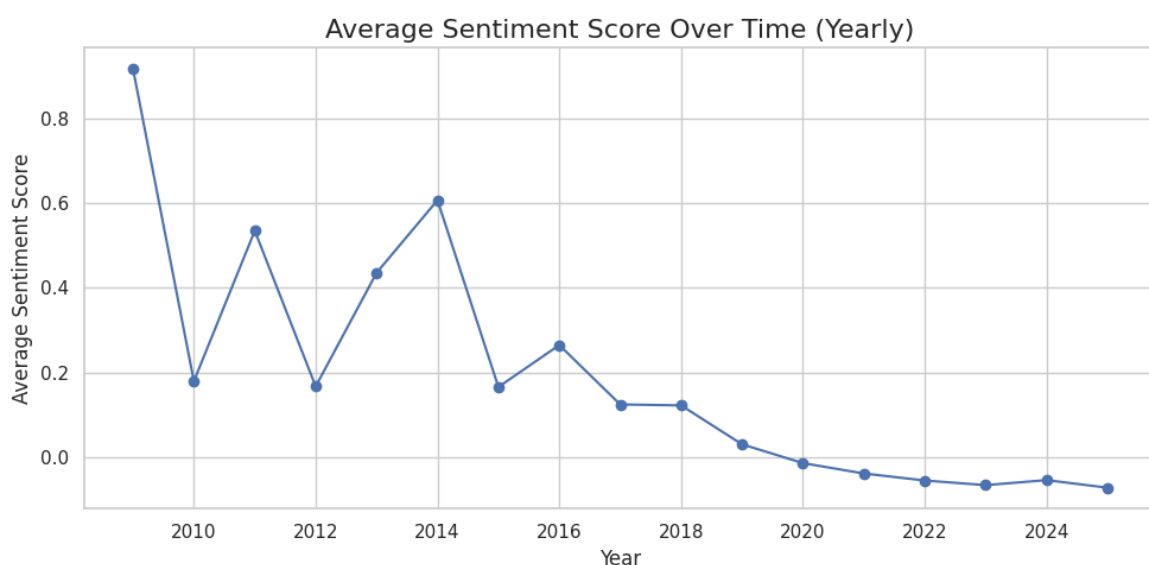


Figure 12: Yearly trend of average sentiment score

Chapter 5: Discussion

In this chapter, the core insights derived from the analysis are examined in relation to the research aims and existing academic discourse. It explores how customers perceive Nike's products and services, the implications of the sentiment-topic relationship, and how these insights compare with existing literature on brand reputation, eWOM, and natural language processing. It also evaluates the outcomes of the sentiment and topic modelling in context and outlines key limitations and directions for further investigation.

5.1 Overview of Key Findings

The research explored how consumers perceive Nike by leveraging natural language processing alongside unsupervised learning methods on reviews collected from Trustpilot. VADER sentiment analysis was employed to categorise more than 4,700 entries into positive, neutral, or negative sentiment groups. Nearly equal proportions of positive and negative sentiments were observed, suggesting a high degree of emotional polarisation in customer feedback.

Word frequency and visual analysis revealed key themes in customer discourse — terms like “order”, “refund”, “customer service”, and “delivery” appeared prominently, indicating that customer experiences with Nike are deeply influenced not just by product quality but also by logistical and service-related factors.

Using Latent Dirichlet Allocation (LDA), topic modelling revealed five core themes discussed within the customer reviews:

1. **Product Quality & Purchase**
2. **Shipping & Delivery**
3. **Customer Support Issues**
4. **Returns & Refunds**
5. **Website/App Experience**

When sentiment classification was cross-referenced with topic allocation, Product Quality & Purchase emerged as the most positively reviewed topic, while Returns & Refunds and Customer Support Issues were dominated by negative sentiment. This suggests that while customers are generally pleased with Nike’s product offerings, there are notable issues in service and post-purchase processes.

The analysis of average sentiment scores per topic further validated these trends, showing that Returns & Refunds had the lowest overall sentiment polarity, while Product Quality had the highest. These insights offer a rich, data-driven understanding of the aspects of Nike’s customer experience that drive brand loyalty versus dissatisfaction.

5.2 Comparison with Existing Literature

This study’s results align with broader patterns observed in sentiment analysis, especially in the application of lexicon-based techniques and the value of integrating topic modelling for contextual interpretation. The VADER tool applied in this research was effective in analysing informal, user-generated text, though some limitations in identifying sarcasm, context shifts, and implicit sentiment were observed — a concern echoed in prior work.

Lexicon-based tools such as VADER are computationally efficient and interpretable but often lack deep contextual understanding when compared to deep learning models (Birjali, Kasri and Beni-Hssane, 2021). This was apparent in the current study when comparing VADER outputs to those from RoBERTa, which captured sentiment nuances more accurately but required significantly more processing power and time.

The importance of hybrid approaches is also supported in recent sentiment analysis literature. Multi-layered pipelines that include data cleaning, sentiment scoring, and thematic modelling are recommended for deeper analytical value (Shayaa *et al.*, 2018). This project followed such a structure, combining VADER sentiment classification with LDA topic modelling to reveal underlying drivers of sentiment. For instance, negative reviews were often linked to issues like refunds and customer service, while positive sentiment centred on product quality.

This method deepens the analysis while providing clearer insights into the emotional factors influencing customer concerns.

5.3 Implications of Nike / Brand Strategy

The findings deliver practical insights that Nike can utilise to enhance customer satisfaction and strengthen brand loyalty. The sentiment-topic analysis revealed a distinct divide between product-related praise and service-related dissatisfaction. These insights can inform targeted improvements in both operational and strategic areas of the business.

The consistently high levels of positive sentiment associated with the Product Quality & Purchase topic highlight Nike's strength in design, fit, and brand appeal. Positive keywords such as "*comfortable*", "*perfect*", and "*great*" reinforce Nike's reputation for delivering products that meet or exceed customer expectations. This suggests that Nike's core product strategy remains effective and could be further amplified through marketing campaigns that highlight real customer satisfaction and product performance.

In contrast, negative sentiment was heavily concentrated in topics such as Returns & Refunds and Customer Support Issues. Customers frequently expressed frustration over delayed refunds, lack of communication, and difficulties contacting support. This indicates that Nike's post-purchase service processes may be undermining overall customer experience. Investing in more transparent return policies, streamlining refund procedures, and improving support responsiveness could mitigate negative feedback and enhance customer trust (Verhoef, Reinartz and Krafft, 2010).

The Shipping & Delivery topic also showed a mix of positive and negative sentiment. While some customers praised fast delivery, others reported delays, missing items, or unclear tracking updates. This inconsistency suggests the need for better logistics coordination and clearer delivery communication.

Additionally, the Website/App Experience topic highlighted technical concerns among users. Reviews referenced checkout issues, app crashes, and navigation difficulties — pointing to weaknesses in Nike's digital infrastructure. These findings align with research highlighting the importance of usability across key stages of the online customer journey, including product identification, cart management, and payment (Mangiaracina, Brugnoli and Perego, 2009). For Nike, improving the speed, reliability, and intuitiveness of its digital platforms could reduce friction and support stronger customer loyalty.

Overall, the findings suggest that while Nike's brand strength is firmly grounded in product excellence, operational touchpoints — especially post-purchase and digital service — require attention. Addressing these areas could not only reduce negative sentiment but also reinforce Nike's positioning as a brand committed to seamless customer experience.

5.4 Strengths and Limitations of the Study

This research demonstrates several strengths in analysing customer sentiment towards Nike. By using real-world, user-generated data from Trustpilot, it captures genuine consumer perspectives. Employing Natural Language Processing (NLP) and topic modelling techniques enabled a more in-depth examination of the textual content, moving beyond simple star ratings to identify underlying themes. Combining sentiment classification with topic identification enabled the detection of emotional tone tied to specific aspects of the customer journey, enhancing the interpretability of the findings for strategic use.

Nevertheless, this study acknowledges certain methodological limitations. While VADER proved effective for analysing short and informal texts, it faces challenges in detecting nuanced expressions such as sarcasm, idioms, and context-sensitive language. Lexicon-based models often oversimplify sentiment polarity and may misclassify ambiguous phrases or intensifiers. These limitations are commonly associated with sentiment analysis tools, many of which face difficulties in capturing emotional subtleties and context effectively (Zucco *et al.*, 2020).

Additionally, the dataset used in this study consisted solely of English-language reviews obtained from Trustpilot. Consequently, the findings may not capture the full extent of Nike's global customer base, particularly in regions where feedback is expressed in different languages or shared on alternative platforms. Social media sites like Twitter and Reddit, for instance, may reflect varied linguistic styles, emotional tones, and feedback structures that differ markedly from those found on review platforms such as Trustpilot. Furthermore, while Latent Dirichlet Allocation (LDA) was suitable for uncovering dominant themes, it assumes word independence and can produce overlapping topics in cases where thematic boundaries are blurred (Boyd-Graber, Hu and Mimno, 2017).

Despite these constraints, the integration of rule-based sentiment analysis with unsupervised topic modelling demonstrates promising potential. This combined approach facilitates scalable and interpretable insights from large volumes of textual data, offering valuable direction for future research aiming to incorporate more diverse datasets or advanced contextual models.

Chapter 6: Conclusion

This dissertation explored public sentiment towards Nike by analysing customer-generated reviews sourced from Trustpilot using natural language processing (NLP) and machine learning techniques. The primary aim was to uncover not only dominant emotional tones within customer feedback but also the underlying thematic drivers. By integrating sentiment classification with topic modelling, the study offered both theoretical insights and practical implications for brand strategy and customer experience management.

The initial chapter presented the rationale for selecting Nike as the case brand, citing its global visibility and active digital footprint. Trustpilot was identified as the primary data source due to its extensive collection of unstructured customer reviews, making it a rich source for opinion mining. Chapter 2 reviewed existing literature on sentiment analysis, NLP tools, and topic modelling within the context of consumer behaviour and brand perception, while also outlining the capabilities and limitations of tools such as VADER and LDA.

Chapter 3 detailed the methodology employed, encompassing data collection and preprocessing, sentiment scoring via the VADER lexicon-based tool, and topic extraction using Latent Dirichlet Allocation (LDA). These tools were selected for their effectiveness in analysing short-form, user-generated content. Chapter 4 presented the analytical results, including visualisations of sentiment distributions, frequently occurring terms, topic summaries, and the relationship between sentiment and topic clusters.

The sentiment analysis showed that feedback was nearly evenly divided between positive and negative reviews, with a smaller proportion classified as neutral. Word frequency plots and word clouds highlighted recurring terms such as "order", "refund", "delivery", and "customer service", indicating common concerns within the review corpus. Topic modelling identified five dominant

themes: Product Quality & Purchase, Shipping & Delivery, Customer Support Issues, Returns & Refunds, and Website/App Experience.

Cross-analysis of sentiment and topic revealed distinct patterns. Product Quality & Purchase was the most positively received topic, suggesting that Nike's core products meet or exceed customer expectations in terms of fit, comfort, and design. In contrast, negative sentiment was concentrated in service-related topics, particularly Returns & Refunds and Customer Support Issues. Many negative reviews referenced issues such as delayed refunds, lack of communication, and unresolved service queries. This division between product satisfaction and service dissatisfaction indicates that while Nike's brand equity remains strong in terms of product value, operational and service elements may be undermining the overall customer experience.

The analysis also highlighted mixed feedback on Shipping & Delivery, where fast fulfilment earned praise but inconsistencies in delivery times and tracking information contributed to dissatisfaction. Website/App Experience emerged as a concern for some users, especially around checkout failures or navigation issues. These findings suggest that Nike's digital platforms could benefit from improved usability, echoing academic research on the importance of seamless user experience in e-commerce environments.

By integrating sentiment analysis and topic modelling, the study was able to move beyond generic rating summaries to deliver context-specific insights. These findings are valuable not only for brand managers and marketers but also for researchers interested in computational social science, customer analytics, and applied NLP.

Answering the Research Questions

This study was designed to address the following key research questions:

1. What are the most common sentiments expressed by Nike customers on Trustpilot?
The dataset contained a near-equal mix of positive and negative sentiment, with clear emotional polarisation.
 2. Which topics or themes emerge most frequently in these reviews?
Five dominant themes were identified: product quality, delivery experience, customer service, returns and refunds, and website/app usability.
 3. How does sentiment vary across different topics and regions?
Sentiment varied significantly by topic. Product-related themes were more positive, while service and support themes carried negative sentiment. Regional patterns were not the focus due to data constraints.
 4. What trends in customer sentiment can be observed over time?
Temporal sentiment trends were not deeply explored in this study due to the dataset's structure, but this represents an area for future research.
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Final Reflections

The present study contributes to the expanding *area* of sentiment mining through the application of accessible NLP tools to brand-specific customer feedback. While the methods used have limitations — particularly in terms of nuance detection and platform scope — they offer a replicable, interpretable framework for analysing extensive unstructured textual data.

For Nike, this research emphasises the need to reinforce its service delivery and post-purchase processes to match the quality of its product offerings. From an academic perspective, the research underscores the value of combining sentiment classification with topic analysis to generate layered insights that move beyond surface-level opinion mining.

Future Work

While this study used interpretable and resource-efficient tools such as VADER and LDA, subsequent studies might investigate the application of deep learning-based architectures such as BERT or RoBERTa, to capture more nuanced sentiment and semantic relationships. These transformer-based models offer superior context awareness but require more powerful computational infrastructure, such as GPU-based machines or cloud platforms. In addition, modern topic modelling techniques like BERTopic or neural topic models could improve topic coherence, especially in short, informal reviews. Extending the dataset to incorporate reviews across diverse languages and digital platforms (e.g., Twitter, Reddit, or Nike's own app) could enhance both the robustness and generalisability of findings.

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